GALGOTIAS UNIVERSITY

ADMISSION NUMBER [redacted]

School of Computer Applications and Technology

Mid Term Examination - Nov 2025

Masters of Computer Applications

Duration : 90 Minutes
Max Marks : 50

Sem I - E1PA107T - Digital Computer Organization

<u>General Instructions</u>

Answer to the specific question asked
Draw neat, labelled diagrams wherever necessary
Approved data hand books are allowed subject to verification by the Invigilator

1)	What is a Decoder? Explain the working of a 2-to-4 line Decoder with a truth table.	K1 **(3)**
2)	Define a Half Adder and a Full Adder. List their number of inputs, number of outputs, and basic functionality.	K1 **(3)**
3)	Compare a Half Adder and a Full Adder in terms of inputs, outputs, and functionality.	K2 **(4)**
4)	Explain the working of an SR Flip-Flop with its truth table and characteristic equation.	K2 **(4)**
5)	Apply the concept of instruction coding to design a 4-bit instruction format that includes an opcode and operand. Explain your design choices.	K3 **(5)**
6)	Describe different types of Micro-operations — Arithmetic, Logic, Shift, and Control — with one example for each.	K3 **(5)**
7)	Analyze the operation of different types of shift registers with suitable diagrams. Compare their data transfer modes and practical applications.	K4 **(8)**
8)	Describe in detail the different modes of shift registers (SISO, SIPO, PISO, PIPO) with diagrams and timing waveforms.	K4 **(8)**
9)	A digital system has several types of registers including general-purpose, data, address, shift, and status registers. a. Classify and describe at least 6 types of registers used in a digital computer system. b. For each register type, design a simple block diagram showing its connections and operation. c. Explain the specific role of each register type in CPU operations. d. Evaluate the impact of using specialized registers on system speed, memory access, and instruction execution efficiency.	K5 **(10)**
	OR	
	Simplify the Boolean function $F(A,B,C,D)=\Sigma(0,1,2,5,8,9,10,14,15)$ using K-Map and implement the simplified function using logic gates.	K5 **(10)**